

WELL COMPLETION REPORT

Synthetic Field Cluster — SYN-Assam-Duliajan (Upper-Assam-type setting)

1. WELL IDENTIFICATION

Well ID	W001
Well Name	SYN-Assam-Duliajan-001
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude	27.371219 N
Longitude	95.331752 E
Well Status	Completed
Well Role	Historical (offset reference well)
Spud Date	09-Dec-2020
Drilling Completed	20-Dec-2020
Report / Completion Date	27-Dec-2020
Total Depth (Driller)	3,470 m
Primary Formation	Sandstone
Formations Penetrated	Claystone / Shale / Sandstone / Limestone
Rig	SYN-RIG-E2000-01
Drilling Days	12 days
Recorded Events	4
Recorded NPT	55.4 hours

2. MACHINE-READABLE HEADER BLOCK

Document Type: Well Completion Report
Well ID: W001
Well Name: SYN-Assam-Duliajan-001
Field: SYN-Assam-Duliajan
Latitude: 27.371219
Longitude: 95.331752
Total Depth: 3,470 m
Primary Formation: Sandstone
Well Status: Completed
Spud Date: 2020-12-09
Event Count: 4
Total NPT: 55.4 hours

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). The well, coordinates, formations, events and measurements described here are generated for software demonstration and are not an operational record of any company. Location names are used for interface readability only.

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3. WELL SUMMARY

3.1 Objective

Well W001 (SYN-Assam-Duliajan-001) was drilled to test the deeper sandstone/limestone sequence and record drilling behaviour across the intermediate shale section. The well was drilled vertically from surface to a driller's total depth of 3,470 m within the SYN-Assam-Duliajan synthetic cluster.

3.2 Drilling Interval

Surface to 3,470 m, completed in 12 rotary drilling days between 09-Dec-2020 and 20-Dec-2020. Average overall penetration rate for the well was 12.0 m/hr on a 24-hour basis.

3.3 Formations Encountered

The well penetrated 4 formation units: Claystone (0–791 m), Shale (791–1,906 m), Sandstone (1,906–3,090 m), Limestone (3,090–3,470 m). The primary objective formation was Sandstone.

3.4 Major Operational Events

A total of 4 reportable events were recorded, comprising 1 × Cementing Issue, 2 × Stuck Pipe, 1 × Mud Loss. Aggregate non-productive time was 55.4 hours. Event detail is given in Section 6.

3.5 Casing Operations

3 casing strings were run: 13-3/8" at 790 m, 9-5/8" at 1,905 m, 7" at 3,470 m. All strings were cemented as detailed in Section 7.

3.6 Completion Status

Well status at report date: **Completed**. The well was completed on the primary sandstone objective and handed over for production testing.

4. OFFSET-WELL CONTEXT

The following synthetic wells of the same cluster lie within short offset distance and share a comparable formation sequence. They are the primary comparison set for offset-well analysis.

Well ID	Well Name	Distance (km)	TD (m)	Primary Formation	Events
W003	SYN-Assam-Duliajan-003	4.68	3,850	Sandstone	8
W002	SYN-Assam-Duliajan-002	4.70	3,450	Sandstone	7
W004	SYN-Assam-Duliajan-004	4.92	3,590	Sandstone	2
W006	SYN-Assam-Duliajan-006	5.84	3,960	Sandstone	0
W009	SYN-Assam-Duliajan-009	7.01	3,180	Sandstone	6
W005	SYN-Assam-Duliajan-005	7.18	3,560	Sandstone	5

5. FORMATION / DEPTH RECORD

Formation tops and bases as recorded from cuttings and log correlation. Depths are driller's depths below rotary table.

Unit	Formation	Top Depth (m)	Bottom Depth (m)	Thickness (m)	Remarks
1	Claystone	0	791	791	Cementing Issue recorded (1 event)
2	Shale	791	1,906	1,115	Stuck Pipe recorded (1 event)
3	Sandstone	1,906	3,090	1,184	Mud Loss, Stuck Pipe recorded (2 events)
4	Limestone	3,090	3,470	380	Routine drilling, no reportable event

Formation: Claystone Top: 0 m Bottom: 791 m
Formation: Shale Top: 791 m Bottom: 1906 m
Formation: Sandstone Top: 1906 m Bottom: 3090 m
Formation: Limestone Top: 3090 m Bottom: 3470 m

6. DRILLING PARAMETERS BY FORMATION

Representative values summarised from the depth-indexed parameter record for this well. Mud weight is reported in ppg, rate of penetration in m/hr, rotary speed in rpm and weight on bit in klbf.

Formation	Interval (m)	Mud Wt (ppg)	MW range	ROP (m/hr)	RPM	WOB (klbf)
Claystone	0–791	9.54	9.0–10.1	32.1	142	8.0
Shale	791–1,906	11.01	10.3–11.8	16.6	126	17.1
Sandstone	1,906–3,090	12.44	11.6–13.3	15.8	113	25.0
Limestone	3,090–3,470	13.89	13.6–14.2	7.3	92	34.9

6.1 Mud Programme Summary

Water-based inhibited mud was used throughout. Mud weight was raised progressively with depth from approximately 9.5 ppg in the upper section to 13.9 ppg at total depth. Mud weight was trimmed to the minimum safe value across the recorded loss intervals and lost-circulation material was maintained in the active system.

6.2 Bit / Hydraulics Note

Rotary speed was held in the 121 rpm range on average with weight on bit averaging 19.7 klbf. Penetration rate declined with depth from 32.1 m/hr in the upper hole to 7.3 m/hr at total depth, consistent with the area trend.

7. DRILLING EVENTS RECORD

All reportable events for this well, in depth order. Each event is recorded against the formation interval containing the event depth.

Event ID	Depth (m)	Formation	Event Type	Severity	Duration (hr)
E0019	92	Claystone	Cementing Issue	High	18.3
E0018	1,370	Shale	Stuck Pipe	Medium	12.8
E0001	2,790	Sandstone	Mud Loss	High	11.9
E0020	3,061	Sandstone	Stuck Pipe	Medium	12.4

Well ID: W001 | Event Type: Cementing Issue | Depth: 92 m | Formation: Claystone | Severity: High | Duration: 18.3 hours

Well ID: W001 | Event Type: Stuck Pipe | Depth: 1370 m | Formation: Shale | Severity: Medium | Duration: 12.8 hours

Well ID: W001 | Event Type: Mud Loss | Depth: 2790 m | Formation: Sandstone | Severity: High | Duration: 11.9 hours

Well ID: W001 | Event Type: Stuck Pipe | Depth: 3061 m | Formation: Sandstone | Severity: Medium | Duration: 12.4 hours

7.1 EVENT DETAIL (1–3 OF 4)

EVENT E0019 — CEMENTING ISSUE AT 92 m (CLAYSTONE)

Well ID: W001
Event ID: E0019
Event Type: Cementing Issue
Depth: 92 m
Formation: Claystone
Severity: High
Duration: 18.3 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Cement slurry losses while cementing across Claystone at 92 m; top of cement lower than designed. Parameters recorded around the event depth: mud weight 9.14 ppg, ROP 34.9 m/hr, 149 rpm, WOB 6.3 klbf.

Response. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.

Outcome. Acceptable bond and annular isolation confirmed after the remedial job. Slurry design and displacement rate revised for the next stage.

Historical observation. Annular coverage is sensitive to losses and centralisation across the shoe interval.

EVENT E0018 — STUCK PIPE AT 1,370 m (SHALE)

Well ID: W001
Event ID: E0018
Event Type: Stuck Pipe
Depth: 1370 m
Formation: Shale
Severity: Medium
Duration: 12.8 hours
Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Drill string became stuck at 1370 m within the Shale section; overpull observed and jarring / working of pipe required. Parameters recorded around the event depth: mud weight 11.07 ppg, ROP 15.8 m/hr, 124 rpm, WOB 17.1 klbf.

Response. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.

Outcome. String freed and circulation restored. Hole conditioned over the affected interval and reaming carried out before drilling ahead.

Historical observation. Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.

EVENT E0001 — MUD LOSS AT 2,790 m (SANDSTONE)

Well ID: W001
Event ID: E0001
Event Type: Mud Loss
Depth: 2790 m
Formation: Sandstone
Severity: High
Duration: 11.9 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Partial loss of drilling fluid recorded at 2790 m in the Sandstone interval; returns reduced and LCM pumped to regain circulation. Parameters recorded around the event depth: mud weight 12.66 ppg, ROP 17.0 m/hr, 107 rpm, WOB 25.6 klbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

7.2 EVENT DETAIL (4–4 OF 4)

EVENT E0020 — STUCK PIPE AT 3,061 m (SANDSTONE)
Well ID: W001 Event ID: E0020 Event Type: Stuck Pipe Depth: 3061 m Formation: Sandstone Severity: Medium Duration: 12.4 hours Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned. Description. Pack-off and stuck pipe in Sandstone at 3061 m; rotation lost, string freed after jarring and lubricant spotting. Parameters recorded around the event depth: mud weight 13.23 ppg, ROP 12.8 m/hr, 100 rpm, WOB 30.8 klbf. Response. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming. Outcome. String freed and circulation restored. Hole conditioned over the affected interval and reaming carried out before drilling ahead. Historical observation. Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.

8. CASING AND CEMENTING RECORD

Casing	Grade / Weight	Shoe Depth (m)	Hole Size	Cement Programme
13-3/8"	K-55, 68 lb/ft	790	17-1/2"	Lead 12.6 ppg / Tail 15.8 ppg, cement to surface
9-5/8"	N-80, 47 lb/ft	1,905	12-1/4"	Lead 12.9 ppg / Tail 15.8 ppg, TOC 670 m
7"	N-80, 29 lb/ft	3,470	8-1/2"	Tail 15.8 ppg, TOC 1,760 m

8.1 Cementing Observations

The following casing/cementing events were recorded and are cross-referenced to Section 7:

- **E0019** — Cementing Issue at 92 m in Claystone (High, 18.3 hr). Cement slurry losses while cementing across Claystone at 92 m; top of cement lower than designed.

Remedial work was carried out as described in Section 7 and annular isolation was confirmed before drilling ahead.

8.2 Pressure Testing

Each string was pressure tested after landing and formation integrity tests were carried out below the 13-3/8" and 9-5/8" shoes. Recorded results were satisfactory except where noted in Section 7.

9. FINAL SUMMARY

Final Depth	3,470 m
Primary Formation	Sandstone
Major Events	2 of 4 events rated High or Critical
Total NPT	55.4 hours
Completion Status	Completed
Offset Risk Classification	HIGH — repeated significant events

9.1 Drilling Challenges

- **Cementing Issue** at 92 m (Claystone) — 18.3 hr cumulative.
- **Stuck Pipe** at 1,370 m, 3,061 m (Shale, Sandstone) — 25.2 hr cumulative.
- **Mud Loss** at 2,790 m (Sandstone) — 11.9 hr cumulative.

9.2 Key Historical Observations

- Annular coverage is sensitive to losses and centralisation across the shoe interval.
- Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.
- Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.
- Formation tops in this well correlate closely with the neighbouring wells listed in Section 4, supporting depth-based offset comparison across the cluster.

9.3 Lessons Learned

- Review slurry design, centralisation and displacement rate for the stage covering 92 m; run a bond log before proceeding.
- Improve hole cleaning and minimise stationary time between 1,370 and 3,061 m; plan a wiper trip before extended surveys in this interval.
- Approach the Sandstone interval (2,790–2,790 m) with LCM pre-mixed and the minimum safe mud weight; monitor returns continuously through this window.

10. DOCUMENT CONTROL

Document Number	WCR-W001-R0
Revision	R0
Issue Date	27-Dec-2020
Prepared By	Drilling Engineering (prototype dataset generator)
Classification	Prototype Training Record — synthetic demonstration data
Source Datasets	wells.csv, well_ formations.csv, drilling_events.csv, drilling_parameters.csv

All identifiers, coordinates, depths, formations, events and measurements in this document are synthetic and were generated for the eRTMAC-NWIS prototype. This document is not an operational record of Oil India Limited or any other operator, and no confidential information is contained in it.